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Genetic aspects of major blood metabolites in the Italian Simmental cattle population

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ABSTRACT

The Simmental breed is widely known for its resilience, robustness, and resistance to disease, and the incidence of ketosis in this breed is therefore generally lower compared with Holsteins. Blood concentrations of nonesterified fatty acids (NEFA), BHB, and urea provide valuable information about the metabolic, health, and nutritional status of lactating animals. In the present study, we estimated h^2 of BHB, NEFA, and urea in blood predicted from milk mid-infrared spectra and assessed their genetic correlation with milk yield and composition traits in the Italian Simmental cattle breed using phenotypes of 3,549 cows in 207 herds. Two sets were considered: early (1,920 records, 1 per cow, between 5 and 35 DIM) and whole (14,378 records, at least 3 per cow, between 5 and 305 DIM) lactation. In early lactation, h^2 estimates were 0.06 for blood BHB as-is, 0.06 for log-transformed BHB, 0.18 for NEFA, 0.14 for log-transformed NEFA, and 0.05 for blood urea. In the whole lactation, the h^2 were 0.09 for blood BHB as-is, 0.16 for log-transformed BHB, 0.03 for NEFA, 0.04 for log-transformed NEFA, and 0.04 for blood urea. As far as the genetic correlations were concerned, blood BHB was positively correlated with NEFA and blood urea. Blood BHB and NEFA were generally positively correlated with milk fat-to-protein ratio and milk, but only the first was negatively correlated with lactose content and positively with SCS. Sires' EBVs for BHB with accuracy ≥0.60 were extrapolated for a posteriori evaluation of the daughters' observed performance. The progeny of the top 5% of sires exhibited, on average, lower blood BHB, NEFA, and urea compared with the daughters of the bottom 5%. Overall, it is highly recommended to monitor the genetic variability of the metabolic traits in the dual-purpose Italian Simmental breed to monitor the incidence of metabolic diseases in future generations.

Key words: ketosis, heritability, metabolic disorder, transition cow

INTRODUCTION

The dual-purpose Simmental cattle breed comprises various populations, such as the Italian Pezzata Rossa and the Austrian Fleckvieh. The Simmental breed is widely known for its resilience, robustness, and resistance to disease (Bronzo et al., 2020), characteristics that are nowadays compromised in specialized dairy breeds. Ketosis is one of the most prevalent diseases in dairy cows (Cainzos et al., 2022); however, the incidence of ketosis in Simmental cows is known to be substantially lower than in Holsteins (Costa et al., 2019). Ederer et al. (2014) reported an average prevalence of 0.60% in the whole Austrian Fleckvieh population. In the same population, Costa et al. (2019) observed percentages from 0.49% to 1.00%, depending on the cows' parity. In specialized dairy breeds, the prevalence of clinical ketosis ranges from 3.70% to 11.60% (Benedet et al. 2019a).

Ketosis is the result of a severe negative energy balance (**NEB**). In early lactation, in fact, dairy cows commonly experience NEB due to the heightened energy demand for milk production, coupled with a reduced feed intake (Grummer et al., 2004). Multiparous cows are more at risk of severe NEB and ketosis, but the severity of the NEB depends on several factors, including the individual cow's productivity level and the feeding strategies during the dry off. To cope with the NEB, a physiological mobilization of body reserves occurs, with the consequent abnormal increase in the blood concentration of nonesterified fatty acids (**NEFA**) and ketone bodies, namely BHB and acetone (Grummer, 1993). Elevated concentration of ketone bodies in blood is defined as hyperketonemia, a condition that can potentially impair immune function, health, disease resistance, and milk production (McArt et

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-24. Nonstandard abbreviations are available in the Notes.

al., 2013). In particular, literature has reported thresholds of BHB and NEFA of greater than 1.20 mmol/L and 0.70 mmol/L, respectively (Ospina et al., 2010; McArt et al., 2013).

During the transition period, Simmental cows undergo various metabolic adaptations, characterized by different energy utilization, inflammatory response, and oxidative patterns compared with Holstein cows (Lopreiato et al., 2019; De Matteis et al., 2021). There is an overall consensus on the superiority of Simmental cows or its crossbreeds in terms of coping with NEB compared with Holsteins, as their mobilization of body reserves is smaller (Knob et al., 2021). Conversely, Strączek et al. (2021) reported that although the reduction in the BCS around the peak of lactation is similar between breeds, Simmental cows demonstrate a more efficient restoration of the BCS compared with Holsteins. Physiological differences between these 2 breeds are reasonably expected. In highly specialized dairy cows, milk synthesis and secretion are the metabolic priorities and are maintained stable at the expense of the metabolic load and equilibrium. Therefore, the NEB tends to be more severe (McCarthy et al., 2010) and the concentrations of NEFA and BHB in the bloodstream greater in the acute phase (Benedet et al., 2020; Catellani et al., 2023).

Beta-hydroxybutyric acid is known to be the most abundant and stable ketone body in cows' biological fluids (Duffield et al., 2009). For phenotyping purpose, recently various authors attempted to develop mid-infrared (**MIR**) spectroscopy prediction models for blood BHB and other traits by using the milk spectra. Although models are generally scarcely or moderately accurate (Benedet et al., 2019b, Luke et al., 2019a), the MIR-predicted blood metabolites still can be considered as useful for screening purposes and for genetic investigations at the population level. Proxies, in fact, have been demonstrated to be efficient tools in animal breeding, and even if the MIR prediction and the reference trait are only moderately correlated, various authors demonstrated that the selection still goes in the desired direction (Bittante et al., 2014; Tiplady et al., 2020, 2022; Costa et al., 2021).

Although the genetic aspects of milk BHB have already been investigated in the Italian Simmental population by Falchi et al. (2021), no studies have explored the phenotypic and genetic aspects of blood indicators related to hyperketonemia in this dual-purpose breed so far. With the present study, we aimed to estimate the genetic parameters in the early and whole lactation of the most popular indicators of NEB, namely blood BHB, NEFA, and urea. For this purpose, phenotypes available consisted of MIR predictions obtained from individual milk spectra. Finally, we estimated their genetic correlations with milk production and composition.

MATERIALS AND METHODS

Milk Data

The initial dataset included 19,657 test-day records and spectral information of individual milk samples of Simmental cows reared in the Veneto region (Italy). Data were retrieved from the official routine milk recording testing archive of the Breeders Association of Veneto Region (**ARAV**, Vicenza, Italy) and covered the period from November 2020 to December 2022. For each test-day record, cow's ID, farm ID, sampling date, DIM, parity, and milk yield (**MY**, kg/d) were available. Fat, protein, casein, and lactose contents (%), milk urea (**$urea_m$**, mg/dL) and milk BHB (**BHB_m**, mmol/L) concentrations, and SCC (cells/mL) were also available. These were determined using the CombiFoss 7 DC machine (Foss, Hillerød, Denmark) of ARAV. The fat-to-protein ratio was calculated using the fat and protein contents, whereas SCC was transformed to SCS to achieve a normal distribution of the data, through the formula of Ali and Shook (1980): $SCS = 3 + \log_2(SCC/100{,}000)$. Finally, following Benedet et al. (2019b), the BHB_m was $\log_{10}$-transformed. Values of milk traits (after transformation, when needed) deviating more than 3 SD from the respective mean were discarded.

Prediction of Blood Traits

For each milk sample, the spectrum containing 1,060 infrared transmittance values within the range of 5,000 and 900 cm^{-1} was used to predict the hematic parameters: blood BHB (**BHB_b**), NEFA, and urea (**$urea_b$**). To develop the model, the MIR spectral points were transformed from transmittance (T) to absorbance (A) through the formula $A = \log_{10}(1/T)$.

The prediction of blood traits was carried out using the same approach described in Benedet et al. (2019b). Briefly, a dataset consisting of 680 blood samples (n. cows = 680) with both milk spectra and reference values was adopted. Concentrations of BHB_b, NEFA and $urea_b$ were determined using the COBAS C501 biochemical analyzer (Roche Diagnostics GmbH, Mannheim, Germany) with Roche BM commercial kits, except for the concentrations of NEFA and BHB, which were measured using an enzymatic colorimetric method (Randox Laboratories Ltd., Ardmore, United Kingdom) at the laboratory of the Experimental Zooprophylactic Institute of the Venezie (Legnaro, Italy).

Due to the non-normal distribution of data points, both BHB_b and NEFA were transformed via $\log_{10}$ (Figure 1), similarly to BHB_m, as already done by Benedet et al. (2019b). On the other hand, $urea_b$ concentration

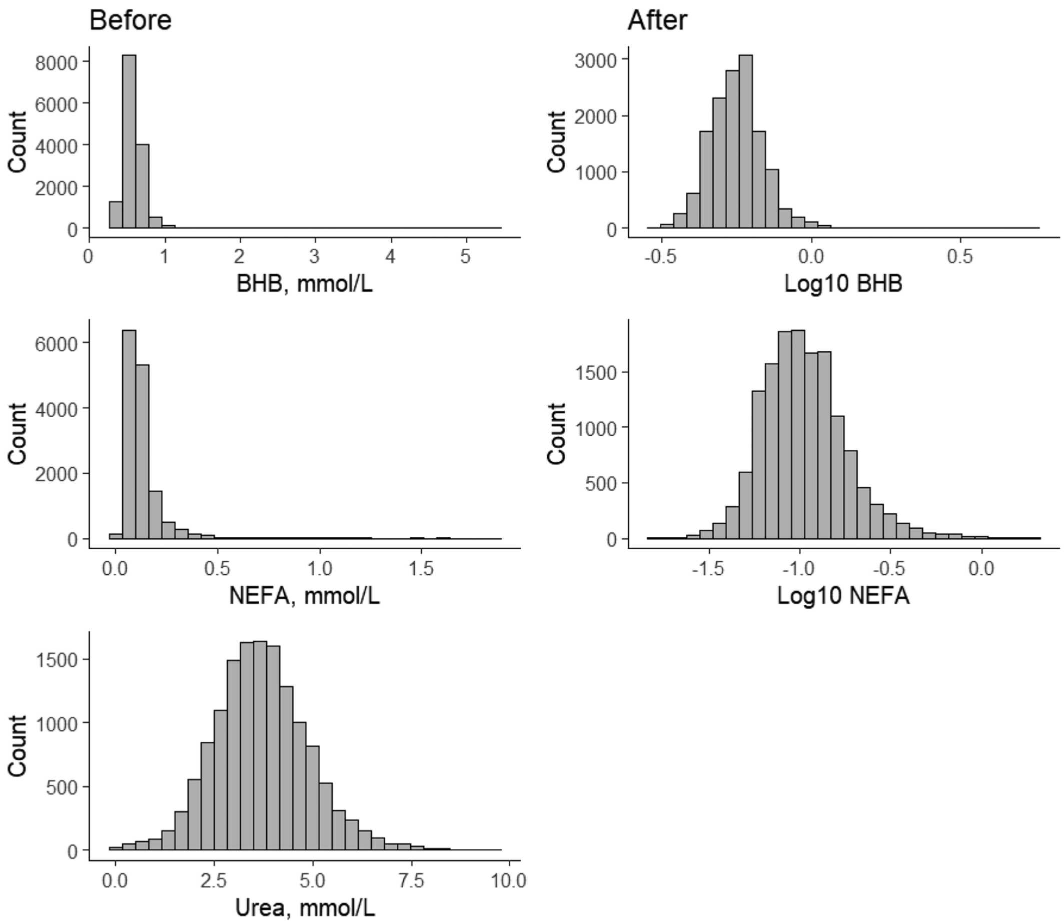

Figure 1. Data points of blood BHB, NEFA, and urea concentration before data editing and removal of outliers. Left side and right side panels show BHB and NEFA values before and after the $\log_{10}$-transformation, respectively.

was used as-is (Figure 1). Other transformations were adopted, such as the root square, but they all resulted in skewed distribution (Supplemental Figure S1, see Notes.).

Of the 680 samples, 476 were used for calibration and development of the prediction equation, whereas the remaining ones (30%, n = 204) were assigned to the validation set. Milk and blood samples belonged to cows of different breeds (including Simmental) across 34 farms in northern Italy during the morning milking, encompassing fresh cows from 5 to 38 DIM in parities from 1 to 10. A partial least squares analysis was carried out using the 'trainControl' function available in the R package 'caret' (Kuhn, 2008). Models were fine-tuned using 10-fold cross-validation repeated 3 times and the number of components was set automatically but not exceeding 20 to avoid overfitting. The final coefficients of determination in cross-validation ($\mathbf{R^2_{CV}}$) were 0.64 for BHB_b, 0.64 for NEFA, and 0.85 for $urea_b$. In external validation, the coefficient ($\mathbf{R^2_V}$) was equal to 0.54, 0.54, and 0.81, respectively.

Models were used in the prediction set with Simmental cows' test-day records (n = 19,657) to obtain the MIR-predicted blood traits, whenever the spectrum was available (see the "Milk Data" section). In the prediction set, values of milk traits and MIR-predicted blood traits deviating more than 3 SD from the respective mean were removed.

Estimation of Genetic Parameters

For the genetic analysis, only test-day records collected up to 305 DIM were kept, and cows with fewer than 3 test-day records within each lactation and herd test-dates with fewer than 3 cows sampled were discarded, ending up with 14,378 test-day records of 3,549 cows reared in 207 herds.

Variance components and h^2 were estimated in the whole dataset ($\mathbf{SET_{5–305}}$) and in a subset containing only data from early lactation ($\mathbf{SET_{5–35}}$), *which is the most critical moment for metabolic diseases in lactating cows*. In particular, only milk records (1 per cow) collected within the first 35 DIM were kept in the $SET_{5–35}$, and again, herds with fewer than 3 cows per year were discarded, finally leading to 1,920 early-lactation records of 1,920 cows reared in 176 herds.

For the genetic analysis, the following repeatability linear animal model was used in the $SET_{5–305}$:

$$y_{ijklm} = \mu + P_i + b(D_{ijklm}) + a_k + c_l + h_m + e_{ijklm}, \quad [1]$$

where y is either the blood or milk phenotype; μ is the overall mean; P is the fixed effect of the ith parity (5 levels, where the last included records from 5 to 10 parity); and D is the fixed covariate of the jth DIM with regression coefficient b. Four random effects were considered: a, which was the animal random effect distributed as $\sim N\left(0, \mathbf{A}\sigma_a^2\right)$, with a pedigree-based relationship matrix ($\mathbf{A}$) and σ_a^2 being the additive genetic variance; c, which was the random effect of the permanent environment (cow) distributed as $\sim N\left(0, \mathbf{I}\sigma_c^2\right)$, with $\mathbf{I}$ being the diagonal identity matrix and σ_c^2 being the variance associated with the cow effect; h, which was the random effect of herd-test-date combination (1,013 levels) distributed as $\sim N\left(0, \mathbf{I}\sigma_h^2\right)$, with $\mathbf{I}$ being the diagonal identity matrix and σ_h^2 being the variance associated with the herd-test-date effect; and e, which was the random residual effect distributed as $\sim N\left(0, \mathbf{I}\sigma_e^2\right)$, with $\mathbf{I}$ being the diagonal identity matrix and σ_e^2 being the variance associated with the residual. To construct $\mathbf{A}$, 3 generations of ancestors from each cow with phenotypes were considered (i.e., 12,836 and 8,215 animals in total for $SET_{5–305}$ and $SET_{5–35}$, respectively).

For the $SET_{5–35}$ (5–35 DIM), the same model [1] was used, but removing the permanent environmental effect c, as only one record per cow was considered. Variance components were computed using the Gibbs sampling approach available in the gibbsf90+ software (Misztal et al., 2014) using a chain with 100,000 samples, discarding the first 10,000 samples as initial burn-in, and saving a sample every 5 iterations. These parameters were chosen based on both previous analyses (Falchi et al., 2021) and on the visual inspection of the chain. The formula used for calculation of h^2 was as follows:

$$SET_{5-305}\text{: } h^2 = \sigma_a^2 \Big/ \left(\sigma_a^2 + \sigma_c^2 + \sigma_h^2 + \sigma_e^2\right);$$

$$SET_{5-35}\text{: } h^2 = \sigma_a^2 \Big/ \left(\sigma_a^2 + \sigma_h^2 + \sigma_e^2\right).$$

The repeatability (t) was computed using repeated measurements of cows only for $SET_{5–305}$, as

$$t = \left(\sigma_a^2 + \sigma_c^2\right) \Big/ \left(\sigma_a^2 + \sigma_c^2 + \sigma_h^2 + \sigma_e^2\right).$$

For the whole lactation ($SET_{5–305}$), based on the estimated variance components for each trait, the EBV were obtained for all the sires in $\mathbf{A}$. Reliability for each EBV was computed from the prediction error variance in blupf90+ (Misztal et al., 2014). The EBV accuracy was then calculated as the square root of the reliability, and only sires with EBV accuracy equal or greater than 0.60 were isolated. Sires were ranked (the top 5% and the bottom 5%) to evaluate the performance of their offspring retrospectively by using a t-test. Pearson phenotypic correlations ($\mathbf{r_p}$) were calculated using raw data in R, whereas pairwise genetic correlations ($\mathbf{r_a}$) were computed with a series of bivariate analyses adopting *model [1] with or without the permanent environmental effect according to the set*.

RESULTS

Descriptive Statistics

An overview of the main descriptive statistics of the blood and milk traits considered in the present study within SET5-35 and SET5-305 is provided in Table 1. In terms of raw mean, MIR-predicted blood BHB_b and NEFA concentrations were higher in early lactation (5–35 DIM) than the whole lactation. On the other hand, $urea_b$ was lower in cows within 35 DIM (Table 1). In the early-lactation set, 2.76% of cows exhibited high BHB_b (>1.20 mmol/L), whereas 4.84% of cows had high NEFA (>0.70 mmol/L). Only 1.46% of cows in early lactation showed both BHB_b and NEFA levels above the literature thresholds. Regarding the whole-lactation set, the percentage of cows exhibiting BHB_b and NEFA above their respective thresholds was lower than 0.50%. The coefficient of variation was the lowest for lactose content (4.2% in early lactation and 4.0% in whole lactation), whereas it was the greatest in the case of NEFA concentration (92.9% and 83.3%, respectively).

Supplemental Table S1 (see Notes) contains the Pearson correlation of blood BHB in 4 different moments within the lactation (i.e., 5–35, 36–100, 101–200, and 201–305 DIM). The association was generally weak or moderate, from 0.17 to 0.44, indicating that this trait is not repeatable within cow and that the variability is high even within animal.

Heritability (h^2)

Regarding blood traits, BHB_b was more heritable in $SET_{5–305}$ than $SET_{5–35}$ (Table 2). In the $SET_{5–305}$, the h^2 of BHB_b was higher in log-transformed (0.16 ± 0.02) than

Table 1. Mean (SD) of the blood[1] and milk traits in the early (5–35 DIM) and whole (5–305 DIM) lactation sets

Trait	$SET_{5–35}$	$SET_{5–305}$
Blood		
BHB, mmol/L	0.62 (0.24)	0.58 (0.15)
Log_{10} BHB	−0.23 (0.14)	−0.25 (0.09)
NEFA, mmol/L	0.28 (0.26)	0.12 (0.10)
Log_{10} NEFA	−0.65 (0.28)	−0.97 (0.23)
Urea, mmol/L	3.15 (1.17)	3.69 (1.21)
Milk		
Yield, kg/d	29.16 (8.07)	25.96 (8.04)
Fat, %	3.92 (0.86)	3.89 (0.71)
Protein, %	3.30 (0.38)	3.42 (0.34)
Fat-to-protein	1.19 (0.27)	1.14 (0.20)
Lactose, %	4.76 (0.20)	4.77 (0.19)
SCS	2.44 (2.13)	2.53 (2.05)
Urea, mg/dL	21.79 (7.10)	24.31 (7.41)
BHB, mmol/L	0.10 (0.06)	0.08 (0.04)
Log_{10} BHB	−1.09 (0.28)	−1.14 (0.25)

[1]Blood traits were predicted via milk mid-infrared spectroscopy.

Table 2. Heritability (SE) of blood[1] and milk traits in the early ($SET_{5–35}$) and whole ($SET_{5–305}$) lactation sets[2]

	$SET_{5–35}$	$SET_{5–305}$	
Trait	h^2	h^2	Repeatability
Blood			
BHB, mmol/L	0.06 (0.04)	0.09 (0.02)	0.19 (0.01)
Log_{10} BHB	0.06 (0.04)	0.16 (0.02)	0.26 (0.01)
NEFA, mmol/L	0.18 (0.05)	0.03 (0.01)	0.03 (0.00)
Log_{10} NEFA	0.14 (0.05)	0.04 (0.01)	0.09 (0.01)
Urea, mmol/L	0.05 (0.03)	0.04 (0.01)	0.09 (0.00)
Milk			
Yield, kg/d	0.05 (0.03)	0.06 (0.02)	0.32 (0.01)
Fat, %	0.31 (0.08)	0.14 (0.03)	0.25 (0.01)
Protein, %	0.25 (0.07)	0.17 (0.03)	0.43 (0.01)
Fat-to-protein	0.27 (0.08)	0.09 (0.02)	0.17 (0.01)
Lactose, %	0.25 (0.08)	0.25 (0.05)	0.47 (0.07)
SCS	0.13 (0.06)	0.17 (0.03)	0.53 (0.01)
Urea, mg/dL	0.13 (0.05)	0.06 (0.01)	0.12 (0.01)
BHB, mmol/L	0.07 (0.04)	0.11 (0.02)	0.21 (0.01)
Log_{10} BHB	0.10 (0.04)	0.14 (0.02)	0.23 (0.01)

[1]Blood traits were predicted via milk mid-infrared spectroscopy.
[2]Repeatability was also calculated for the whole set.

as-is (0.09 ± 0.02); in the $SET_{5–35}$ the h^2 of BHB_b had the same value if as-is or log-transformed (0.06 ± 0.04). On the other hand, NEFA was more heritable in $SET_{5–35}$ than $SET_{5–305}$ (Table 2). The h^2 in $SET_{5–35}$ were 0.18 ± 0.05 and 0.14 ± 0.05 for NEFA and log-transformed NEFA, respectively. In $SET_{5–305}$, NEFA was the least heritable blood trait with h^2 of 0.03 ± 0.01 and 0.04 ± 0.01. The h^2 of $urea_b$ was very low (0.05 ± 0.03 in $SET_{5–35}$; 0.04 ± 0.01 in $SET_{5–305}$). The repeatability for blood traits in $SET_{5–305}$ ranged from 0.03 (NEFA as-is) to 0.26 (log-transformed BHB_b). Regarding the milk traits, the h^2 of milk traits ranged from 0.05 (MY) to 0.31 (fat content) in $SET_{5–35}$, whereas h^2 of milk components in the $SET_{5–305}$ ranged from 0.06 (MY and $urea_m$ concentration) to 0.25 (lactose content; Table 2). The repeatability in $SET_{5–305}$ ranged from 0.12 ($urea_m$) to 0.47 (lactose content).

Correlations

Blood BHB_b and NEFA were positively correlated in $SET_{5–35}$ and $SET_{5–305}$, both at the phenotypic and genetic levels (Tables 3 and 4). However, the magnitude of the association was greater in $SET_{5–35}$ than in $SET_{5–305}$. The r_p between log-transformed BHB_b and NEFA were 0.46 in $SET_{5–35}$ and 0.16 in $SET_{5–305}$ (Table 3). Correspondingly, the r_a were 0.65 and 0.23, respectively (Table 4).

The r_p of urea with log-transformed BHB_b and NEFA were negative and weak. On the other hand, the genetic correlation was positive and was stronger in $SET_{5–35}$ (0.45) than in $SET_{5–305}$ (0.18 and 0.17; Table 4).

Milk yield was very weakly associated with BHB_b and NEFA at phenotypic level (Table 3), however, on the other hand, its r_a was positive and ranged from 0.09 (BHB_b as-is) to 0.49 (NEFA as-is) in $SET_{5–35}$ and from 0.17 (BHB_b as-is) to 0.19 (log-transformed BHB_b) in $SET_{5–305}$ (Table 4). The genetic correlation of MY with urea was negative and stronger in $SET_{5–35}$ (−0.70) than in $SET_{5–305}$ (−0.20, Table 4).

At the genetic level, fat in milk was positively and strongly associated with BHB_b and NEFA in early lactation, whereas these correlations in the whole lactation were negative and close to zero (Table 4). Conversely, genetic correlations between protein in milk and BHB_b and NEFA were always negative in both sets (early and whole lactation; Table 4).

In $SET_{5–35}$, the correlations of BHB_b and NEFA with fat-to-protein were stronger, ranging from 0.65 (log-transformed NEFA) to 0.94 (BHB_b as-is). On the other hand, at phenotypic level correlations, although positive, were lower (Table 3). The r_a between fat-to-protein and urea was stronger in $SET_{5–305}$ (0.39) than in $SET_{5–35}$ (0.22; Table 4).

In $SET_{5–35}$, at the genetic level, lactose content was negatively correlated with BHB_b and NEFA (both as-is and log-transformed) and ranged from −0.31 (NEFA as-is) to −0.70 (log-transformed BHB_b; Table 4). On the other hand, the correlation between SCS and blood traits in $SET_{5–35}$ was positive, with the strongest correlation being with urea (0.60; Table 4).

As expected, urea concentrations in blood and milk were positively correlated: at the genetic level, the correlation was stronger in $SET_{5–35}$ (0.91) than in $SET_{5–305}$ (0.39; Table 4). The correlations between BHB_m and BHB_b (both log-transformed and as-is) were positive and strong. The strongest r_p was found between BHB_m and BHB_b as-is, both in $SET_{5–35}$ (0.76) and in $SET_{5–305}$ (0.70; Table 3). On the other hand, the strongest r_a was between BHB_m as-is and BHB_b log-transformed in $SET_{5–35}$ (0.95)

Table 3. Pearson correlations between blood[1] and milk traits in the early lactation ($SET_{5–35}$, above diagonal) and whole lactation ($SET_{5–305}$, below diagonal)

	Blood					Milk								
Trait	BHB	Log_{10} BHB	NEFA	Log_{10} NEFA	Urea	Yield	Fat	Protein	Fat-to-protein	Lactose	SCS	Urea	BHB	Log_{10} BHB
Blood														
BHB, mmol/L		0.97***	0.46***	0.48***	−0.09**	*NS*	0.42***	−0.22***	0.53***	−0.12***	0.04**	0.05**	0.76***	0.65***
Log_{10} BHB	0.94***		0.42***	0.46***	−0.09**	0.08**	0.40***	−0.27***	0.53***	−0.11***	0.03**	0.04**	0.74***	0.70***
NEFA, mmol/L	0.30***	0.24***		0.85***	−0.12**	*NS*	0.38***	0.07***	0.33***	−0.23***	0.13***	0.17***	0.37***	0.30***
Log10 NEFA	0.18***	0.16***	0.87***		−0.10**	*NS*	0.36***	*NS*	0.34***	−0.22***	0.14***	0.20***	0.44***	0.40***
Urea, mmol/L	−0.04***	−0.06***	−0.14***	−0.16***		0.05*	*NS*	*NS*	*NS*	*NS*	*NS*	0.67***	*NS*	*NS*
Milk														
Yield, kg/d	−0.02***	−0.02***	0.06***	0.05***	*NS*		−0.08***	−0.18***	*NS*	0.13***	−0.22***	0.06**	*NS*	*NS*
Fat, %	0.21***	0.20***	0.10**	0.02***	*NS*	−0.21***		0.22***	0.86***	−0.24***	0.15***	0.16***	0.23***	0.17***
Protein, %	−0.21***	−0.24***	−0.22***	−0.31***	0.06***	−0.23***	0.33***		−0.28***	−0.27***	0.17***	*NS*	−0.14***	−0.20***
Fat-to-protein	0.34***	0.35***	0.25***	0.21***	*NS*	−0.09***	0.84***	−0.23***		−0.10***	0.06***	0.17***	0.29***	0.26***
Lactose, %	−0.05***	−0.05***	−0.05***	−0.04***	*NS*	0.19***	−0.13***	−0.07***	−0.09***		−0.46***	*NS*	−0.17***	−0.13***
SCS	*NS*	*NS*	0.03**	0.03**	0.02**	−0.16***	0.12***	0.13***	0.05***	−0.38***		*NS*	0.08***	0.08***
Urea, mg/dL	−0.07***	−0.09***	0.04*	0.06***	0.77***	0.03**	0.12***	0.10***	0.07***	*NS*	0.03***		0.00***	0.01*
BHB, mmol/L	0.70***	0.69***	0.38***	0.39***	*NS*	*NS*	0.05***	−0.27***	0.20***	−0.09***	0.03***	−0.12***		0.89***
Log_{10} BHB	0.56***	0.63***	0.28***	0.33***	*NS*	*NS*	*NS*	−0.28***	0.16***	−0.08***	0.02***	−0.12***	0.91***	

[1]Blood traits were predicted via milk mid-infrared spectroscopy.

***$P < 0.001$; **$P < 0.01$; *$P < 0.05$; *NS* = not significant.

Table 4. Genetic correlations (SE) between blood[1] and milk traits in the early lactation ($SET_{5–35}$, above diagonal) and whole lactation ($SET_{5–305}$, below diagonal)

	Blood					Milk								
Trait	BHB	Log_{10} BHB	NEFA	Log_{10} NEFA	Urea	Yield	Fat	Protein	Fat-to-protein	Lactose	SCS	Urea	BHB	Log_{10} BHB
Blood														
BHB, mmol/L		0.99 (0.01)	0.64 (0.22)	0.64 (0.22)	0.38 (0.54)	0.09 (0.42)	0.91 (0.10)	−0.75 (0.29)	0.94 (0.07)	−0.45 (0.26)	0.39 (0.44)	0.40 (0.43)	0.88 (0.13)	0.77 (0.18)
Log_{10} BHB	1.00 (0.00)		0.65 (0.20)	0.65 (0.20)	0.45 (0.30)	0.13 (0.37)	0.82 (0.12)	−0.11 (0.45)	0.83 (0.15)	−0.70 (0.28)	0.27 (0.71)	−0.09 (0.3)	0.95 (0.05)	0.80 (0.14)
NEFA, mmol/L	0.07 (0.16)	0.19 (0.15)		0.98 (0.02)	0.16 (0.44)	0.49 (0.23)	0.71 (0.13)	−0.32 (0.25)	0.85 (0.13)	−0.31 (0.19)	0.00 (0.29)	0.39 (0.34)	0.50 (0.24)	0.35 (0.24)
Log_{10} NEFA	0.27 (0.14)	0.23 (0.13)	0.98 (0.02)		0.45 (0.30)	0.42 (0.29)	0.63 (0.15)	−0.22 (0.23)	0.65 (0.18)	−0.56 (0.2)	0.16 (0.26)	0.56 (0.21)	0.36 (0.32)	0.35 (0.25)
Urea, mmol/L	0.32 (0.14)	0.18 (0.14)	0.31 (0.17)	0.17 (0.17)		−0.70 (0.25)	0.42 (0.34)	0.04 (0.41)	0.22 (0.26)	−0.12 (0.37)	0.6 (0.32)	0.91 (0.06)	0.36 (0.55)	−0.22 (0.52)
Milk														
Yield, kg/d	0.17 (0.17)	0.19 (0.14)	0.18 (0.20)	0.17 (0.18)	−0.20 (0.20)		0.26 (0.38)	−0.04 (0.29)	0.05 (0.38)	−0.16 (0.42)	−0.42 (0.32)	−0.52 (0.24)	0.06 (0.35)	−0.07 (0.32)
Fat, %	−0.06 (0.14)	−0.09 (0.12)	−0.07 (0.16)	−0.23 (0.13)	0.27 (0.13)	−0.35 (0.17)		0.35 (0.18)	0.94 (0.03)	−0.26 (0.21)	−0.08 (0.28)	0.58 (0.17)	0.69 (0.22)	0.47 (0.24)
Protein, %	−0.38 (0.12)	−0.33 (0.11)	−0.41 (0.13)	−0.50 (0.11)	−0.26 (0.18)	0.02 (0.23)	0.62 (0.09)		−0.14 (0.19)	0.47 (0.26)	−0.02 (0.30)	−0.14 (0.28)	−0.70 (0.33)	−0.33 (0.27)
Fat-to-protein	0.18 (0.14)	0.13 (0.13)	0.11 (0.17)	0.06 (0.14)	0.39 (0.14)	−0.40 (0.17)	0.89 (0.04)	0.02 (0.16)		−0.52 (0.24)	−0.13 (0.28)	0.67 (0.20)	0.89 (0.13)	0.68 (0.24)
Lactose, %	0.03 (0.12)	−0.02 (0.11)	−0.34 (0.15)	−0.31 (0.12)	0.21 (0.12)	−0.19 (0.17)	0.07 (0.11)	0.20 (0.12)	−0.07 (0.13)		−0.65 (0.20)	−0.03 (0.23)	−0.43 (0.3)	−0.51 (0.24)
SCS	−0.06 (0.14)	0.00 (0.13)	−0.15 (0.21)	0.08 (0.14)	0.14 (0.16)	−0.07 (0.16)	−0.21 (0.14)	−0.35 (0.13)	−0.06 (0.13)	−0.27 (0.11)		0.43 (0.24)	0.52 (0.31)	0.35 (0.33)
Urea, mg/dL	−0.04 (0.14)	−0.02 (0.12)	0.07 (0.19)	0.01 (0.14)	0.75 (0.06)	−0.17 (0.22)	0.38 (0.12)	0.02 (0.13)	0.42 (0.12)	0.16 (0.10)	0.15 (0.14)		−0.18 (0.34)	−0.50 (0.24)
BHB, mmol/L	0.79 (0.06)	0.77 (0.05)	0.15 (0.16)	0.35 (0.13)	0.05 (0.13)	0.25 (0.16)	−0.11 (0.13)	−0.44 (0.10)	0.15 (0.14)	−0.14 (0.11)	0.11 (0.13)	−0.27 (0.12)		0.96 (0.05)
Log_{10} BHB	0.83 (0.06)	0.74 (0.06)	0.22 (0.15)	0.33 (0.13)	−0.03 (0.13)	0.25 (0.16)	−0.13 (0.13)	−0.35 (0.10)	0.07 (0.13)	−0.17 (0.11)	0.12 (0.13)	−0.28 (0.12)	0.99 (0.01)	

[1]Blood traits were predicted via milk mid-infrared spectroscopy.

and between BHB_m log-transformed and BHB_b as-is in $SET_{5–305}$ (0.83; Table 4).

Performance of the Offspring of Top and Bottom Sires

In this study, the EBVs of log-transformed BHBb were used to rank sires and to compare the performance of their oCspring retrospectively. In this particular case, the top sires were those with the lowest EBVs and, vice versa, the bottom those with the highest EBVs. The BHB_b levels of the daughters from the top 5% of sires averaged 0.526 ± 0.004 mmol/L which was significantly lower ($P \leq 0.001$) than those of the daughters from the bottom 5% of sires (0.603 ± 0.006 mmol/L, Figure 2). Additionally, NEFA and $urea_b$ concentrations of the daughters from the top 5% of sires (0.124 ± 0.003 mmol/L and 3.58 ± 0.04, respectively) were lower (but not significantly) than those of the daughters from the bottom 5% of sires (0.133 ± 0.005 mmol/L and 3.65 ± 0.07, respectively). Moreover, BHB_m levels of the daughters from the top 5% of sires averaged 0.07 ± 0.04 mmol/L which was significantly lower ($P < 0.001$) than those of the daughters from the bottom 5% of sires (0.09 ± 0.04 mmol/L, Figure 2). Indeed, the bull's EBV for blood $\log_{10}$ BHB was positively correlated with the bull's EBV for milk $\log_{10}$ BHB (Figure 3).

DISCUSSION

The aim of the present study was to determine genetic parameters of blood BHBb, NEFA, and urea concentration predicted from milk infrared spectra in Italian Simmental cattle breed. In fact, comparing the phenotypic and genetic variability of these traits in early lactation versus the entire lactation allows to evaluate how different can be the interpretation of a trait measured in the first 35 DIM than in other moment of the lactation. The same can be said for the covariance between traits.

To make discussion fair and possible, and to compare the results with those reported by different authors, multiple forms of the same trait were considered in the genetic analyses (i.e., BHB_b as-is and log-transformed BHB_b and NEFA as-is and log-transformed NEFA). In some studies, the BHB_b could be reasonably normally and therefore, not log-transformed for genetic analysis (Belay et al., 2017). In other cases, other types of transformation were done, e.g., the root square transformation in NEFA (Luke et al., 2019b). Deciding whether to conduct some type of trait transformation depends on the characteristics of the starting dataset. However, the distribution of health traits and blood biomarkers is often skewed, with clear implications for their linear applications, as there may be different interpretations for their mean and SD. Figure 1 shows the distribution of predicted BHB_b and NEFA in the 2 forms, i.e., as-is and log-transformed, in addition to the as-is urea concentration.

Regarding phenotype quality, the equations developed in this study provided predictions of different accuracy and interpretation. Considering the classification of Williams (2004), when the coefficient of determination of an infrared prediction model is between 0.50 and 0.65, the predicted trait should be exclusively used for detection of extreme values or comparison of groups. Instead, values in the range of 0.66 to 0.81, 0.82 to 0.90 and ≥0.91 can be considered as accurate enough for approximate screening, good quantitative screening, and punctual prediction, respectively. Grelet et al. (2021) proposed a slightly different classification (based on the R^2_{CV} and other model performance) according to which the MIR predictions obtained with a R^2_{CV} of (1) ≥0.98 can be used for any application, (2) between 0.94 and 0.97 for quality control, (3) between 0.89 and 0.93 for quantitative screening, (4) between 0.74 and 0.89 for rough screening, and (5) between 0.55 and 0.74 for discrimination. Based on the authors, then, models with R^2_{CV} in the range of 0.55 to 0.74 should be considered imprecise and only able to detect very extreme values (Grelet et al., 2021). None of these studies provided a classification for model performance in external validation, i.e., considering the R^2_V. However, based on their ranges for the cross-validation, we are confident that our MIR-predicted BHB_b, NEFA, and $urea_b$ can be considered as reliable and robust for our purpose, i.e., large-scale investigation, genetic parameters estimation, and correct identification of superior animals in the population. In this regard, studies have demonstrated that proxies are efficient tools for animal breeding, and even if the MIR prediction and the reference trait are only moderately correlated, the response to selection exists and goes in the desired direction (Bittante et al., 2014;Tiplady et al., 2020, 2022; Costa et al., 2021).

Descriptive Statistics

The descriptive statistics of BHB_b of this study were consistent with those reported by authors investigating genetic aspects of ketosis or hyperketonemia in Holstein cows in early lactation (Luke et al., 2019b; Benedet et al., 2020). For the whole lactation (5–305 DIM), Lou et al. (2022) reported an average BHB_b of 1.45 mmol/L calculated using MIR predictions in 11,609 Holstein cows. Belay et al. (2017) reported an average of 1.19 mmol/L for the same trait in 324,920 Norwegian Red cows sampled from 11 to 120 DIM.

The raw mean of NEFA in Simmental cows in $SET_{5–35}$ (0.28 mmol/L) was lower than the values reported by other studies that considered Holstein cows in early lac-

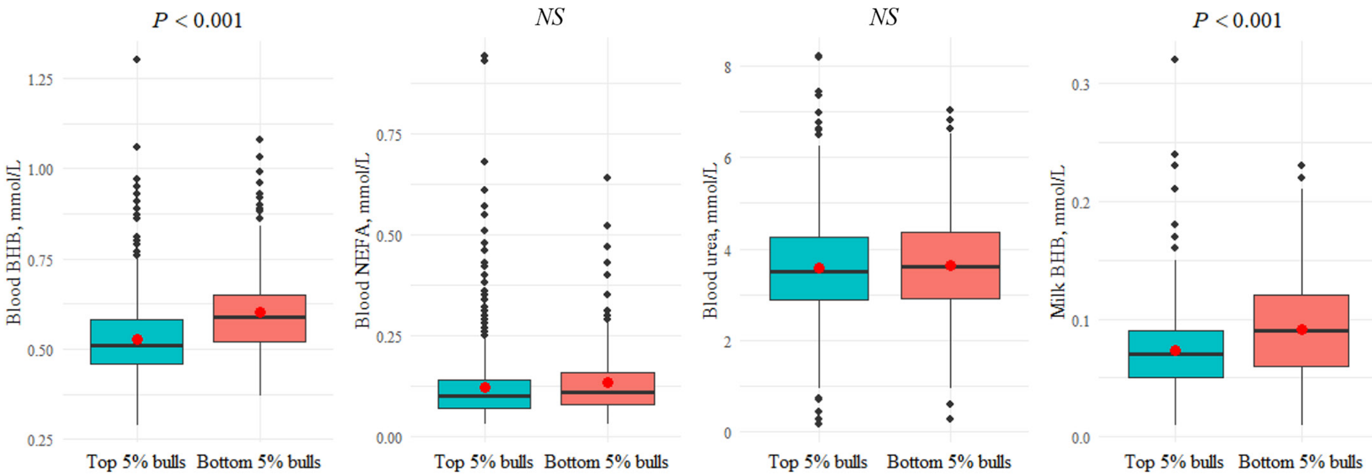

Figure 2. Performance and significance (*NS* = not significant) in the whole lactation ($SET_{5–305}$) of the daughters of the top (725 test-day records, 177 cows) and bottom (285 test-day records, 69 cows) 5% of sires for blood $\log_{10}$ BHB. Each black dot represents an outlier value, whereas red dots represent the mean. Lower and upper edges of the boxes represent the first (Q1) and third (Q1) quartiles, respectively. Midlines represent the median values; whiskers represent 1.5 times the interquartile range (Q3–Q1).

tation (Luke et al., 2019b; Benedet et al., 2020). Benedet et al. (2020) observed a raw mean of predicted NEFA of 0.41 mmol/L in 22,718 test-day records of 13,106 Holstein cows in 456 herds. Luke et al. (2019b) reported a raw mean of measured NEFA of 0.55 mmol/L in 1,393 cows located on 14 farms in southeastern Australia. Benedet et al. (2020), who investigated the effects, including breed, on predicted blood BHB_b and NEFA in early lactation, reported that Simmental cows had lower BHB_b and NEFA concentrations compared with Holstein cows but higher concentrations than Brown Swiss cows. The lower values found for the indicators of inflammatory status in Simmental compared with Holsteins could be due to the lower MY of the former breed.

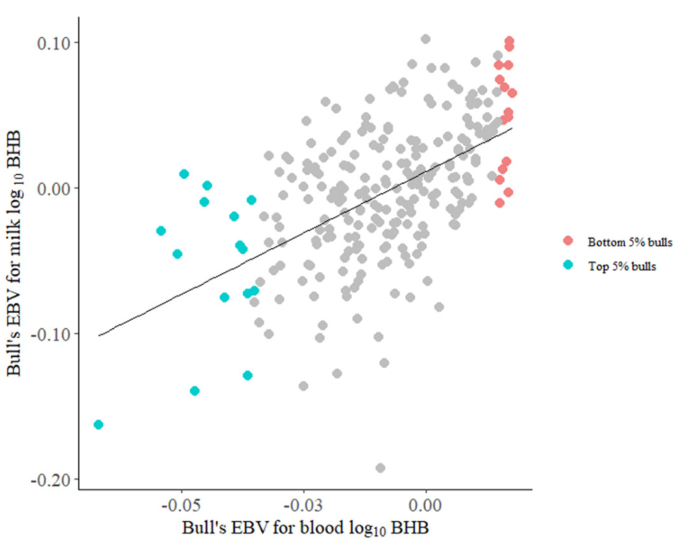

Figure 3. Correlation between blood $\log_{10}$ BHB and milk $\log_{10}$ BHB using the EBV of the top 5% and bottom 5% sires.

The raw mean of $urea_b$ concentration in our study was lower than the values reported by Luke et al. (2019b) in Holstein cows, who used reference blood values instead of MIR predictions. Lopreiato et al. (2019) observed higher concentration of $urea_b$ in Simmental than Holstein cows in the peripartum period. This finding was also confirmed by the study of Benedet et al. (2020), which observed that blood urea nitrogen in the Simmental cows was higher than in the Holstein cows. The raw means of BHB_m in Simmental cows in our study (both in $SET_{5–35}$ and $SET_{5–305}$) were consistent with the values observed in the study of Falchi et al. (2021), who reported a raw mean of BHB_m ranging from 0.06 (year of birth 2009) to 0.09 (year of birth 2018) in 30,461 Simmental cows within the first 90 d of lactation.

Heritability (h^2)

Several studies have evaluated the genetic factors of ketosis-related blood metabolites in the Holstein breed (Luke et al., 2019b; Benedet et al., 2020; Lou et al., 2022). However, to our knowledge, no study has evaluated the h^2 of these traits in the Simmental breed. As reported in the literature, during the transition period Simmental cows undergo different metabolic adaptations, characterized by variations in energy use, inflammatory response, and oxidative models compared with Holstein cows (Lopreiato et al., 2019; De Matteis et al., 2021).

The h^2 of BHB_b in early lactation in Simmental cows was lower than that reported in other studies in which other breeds were considered (Belay et al., 2017; Benedet et al., 2020). In particular, Benedet et al. (2020) reported h^2 of 0.21 for predicted log-transformed BHB_b in Italian Holstein cows in the first 35 DIM. Belay et al. (2017) reported h^2 of 0.25 for BHB_b as-is in 324,920 Norwegian Red cows in the first stage of lactation (11–30 DIM) reared in 3,539 herds. On the other hand, Luke et al. (2019b) reported h^2 of 0.09 for BHB_b measured and log-transformed in Holstein cows in early lactation, which is consistent with our results. The h^2 of BHB_b (as-is and log-transformed) in $SET_{5–305}$ was consistent with the results of Lou et al. (2022), who found an h^2 of 0.12 in predicted and as-is BHB_b using data from 12,157 Holstein cows throughout the whole lactation (5–305 DIM).

The h^2 of NEFA in early lactation was consistent with the findings reported for Holstein cows (Luke et al., 2019b; Benedet et al., 2020). In particular, Luke et al. (2019b) reported an h^2 of 0.18 for NEFA with square root transformation, and Benedet et al. (2020) observed an h^2 of 0.14 for NEFA as-is. The h^2 values estimated for $urea_b$ were low in both early and whole lactations. However, these values were lower than that (0.18) reported by Luke et al. (2019b), who used reference blood values instead of MIR predictions.

The estimated h^2 of milk composition traits agreed with previous findings in Simmental or Fleckvieh cows in other studies (Costa et al., 2019; Falchi et al., 2021; Ablondi et al., 2023). Regarding the MY, h^2 estimates are low when compared with other breeds such as Holstein but not far from other estimates reported in literature for Simmentals. In particular, h^2 estimates for daily MY in Italian Simmental by Ablondi et al. (2023; 0.09) and Falchi et al. (2021; 0.10) are just a couple of points larger than the h^2 reported in the present study. In general, differences in terms of h^2 across studies depend on the country or population considered, the size and restriction of the data, the pedigree depth, and the model adopted. As an example, in Falchi et al. (2021) the h^2 of BHB_m was estimated at 0.09 using data of 30,461 Simmental cows in the first 90 DIM. Regarding BHB_m, Lee et al. (2016) found h^2 equal to 0.08, 0.11, and 0.09 in first-, second-, and third-lactation Holstein cows in whole lactation.

Correlations

The positive r_a between BHB_b and NEFA in early lactation was consistent with that found by Benedet et al. (2020) using MIR-predicted traits in Italian Holsteins, i.e., r_a of 0.51 ± 0.05 between log-transformed BHB_b and NEFA. In addition, the same authors estimated the r_a between the same traits in different moments during the first 35 DIM, the greatest association was found in the window of 5 to 10 DIM (0.78 ± 0.06; Benedet et al., 2020). Conversely, Luke et al. (2019b) reported a lower r_a between these traits. In this study, the magnitude of the association, however, was greater in the $SET_{5–35}$ than $SET_{5–305}$, supporting the fact that indicators fluctuations are informative of the same status. In fact, metabolic disorders majorly occur in the first DIM.

At genetic level, $urea_b$ was positively correlated with both BHB_b and NEFA. Conversely, Luke et al. (2019b) found a negative association (−0.17) between $urea_b$ and NEFA in Holstein cows sampled within 30 DIM. Although high-producing breeds (i.e., Holstein) are genetically predisposed to mobilize more lipids than muscle reserves during NEB, dual-purpose cows such as Simmental are probably more inclined to derive energy from both lipids and proteins present in muscle tissue when there is an increased energy demand.

The positive associations at the genetic level between MY and BHB_b and NEFA were consistent with other studies on other cattle breeds (Belay et al., 2017; Benedet et al., 2020). Benedet et al. (2020) considered Holstein cows in the first 35 DIM of lactation and reported a stronger correlation between MY and NEFA (0.53) than MY and BHB_b (0.22), as found in this study in Simmental cows in the same stage of lactation. Belay et al. (2017) investigated the genetic associations of BHB_b with milk production traits in Norwegian Red cows in different stages of lactation and reported a correlation of 0.20 between 11 and 30 DIM and 0.10 between 11 and 120 DIM.

As reported in other breeds (Belay et al., 2017; Benedet et al., 2020), protein content in milk was negatively correlated with BHB_b, as well as NEFA, both in early lactation and the whole lactation. On the other hand, the r_a found in Simmental cows in this study between milk fat and BHB_b and NEFA were stronger than those reported in other studies focused on other breeds that reported correlation close to zero (Belay et al., 2017; Benedet et al., 2020).

The fat-to-protein ratio was positively correlated with BHB_b, NEFA, and $urea_b$ in both early lactation and the whole lactation, confirming this trait to be a potential proxy of NEB in dairy cows. The r_a between this ratio and BHB_b was stronger in early lactation and greater than the value of 0.25 reported by Benedet et al. (2020). On the other hand, the r_a between this ratio and $urea_b$ was stronger in the whole lactation than in early lactation (Table 4).

In line with findings by Benedet et al. (2020) for Holstein cows, lactose content in milk negatively correlated with BHB_b and NEFA in early lactation. Additionally, Belay et al. (2017) reported a negative genetic association ranging from −0.23 (11–30 DIM) to −0.15 (91–120 DIM) between lactose content and BHB_b in Norwegian Red cows. Costa et al. (2019) found an inverse relationship at genetic level between lactose content and ketosis in Austrian Fleckvieh.

Although the BHB_m concentration predicted from the milk spectra does not perfectly mirror the hematic concentration due to its fluctuation during the day, the r_p and r_a between BHB_m and BHB_b were positive and strong both in early lactation and the whole lactation. The Milkoscan manufacturers (Foss, Hillerød, Denmark), in fact, proposed BHB_m as an indicator of subclinical ketosis at herd level (i.e., bulk milk) rather than at individual level (Renaud et al., 2019). In early lactation, BHB_b (as-is and log-transformed), was positively genetically associated with SCS. This somewhat corroborates the findings of other studies that reported a positive r_a between ketosis and mastitis (Pryce et al., 2016; Costa et al., 2019). In particular, Costa et al. (2019) reported a positive r_a (0.35) between these 2 diseases in 84,289 Fleckvieh cows in the first 150 DIM.

General Perspective

Even though the MIR model accuracy is far from being considered as optimal, the use of predicted blood phenotypes for selection purposes is meaningful, as they at least allow for identification of best and worst sires and dams (Grelet et al., 2021). Based on Cecchinato et al. (2009) and Costa et al. (2021), it is frequent that phenotypes predicted via infrared spectroscopy show strong a genetic correlation (scarce or negligible animals' reranking) with the reference values even though the phenotypic correlation is moderate or weak.

Our retrospective evaluation revealed that offspring of the top 5% bulls had significantly lower blood BHB levels compared with those of the bottom 5% (Figure 1). The incidence of ketosis in the Simmental breed is known to be generally low, as the genetic pressure on milk productivity in the past was low to moderate, with less risk of adverse effects on health and fitness over successive generations (Oltenacu and Broom, 2010). Conversely, the current breeding goal for the Simmental breed in Italy includes various characteristics and emphasizes dual-purpose features (https://anapri.eu/it/indice-idas.html; ANAPRI, 2024). The aim of the breeders' association is to achieve a sustainable improvement in the profitability of milk production while maintaining a balanced consideration of beef production and, above all, fitness characteristics (ANAPRI, 2024; Genetic Austria, 2024). In the Austrian and Italian breeding programs, fitness characteristics have a significant weight, i.e., 44% in the Austrian Fleckvieh and 47% in the Italian Pezzata Rossa population, followed by milk traits (38% and 33%, respectively) and meat characteristics (18% and 20%, respectively). Regarding milk, the emphasis is not placed on milk production per se, but indirectly on the yield of fat or protein (ANAPRI, 2024; Genetic Austria, 2024). However, it is advisable to explore the genetic characteristics of the metabolic blood traits, even in the Simmental breed, to monitor metabolic diseases within the population and maintain a low incidence of impactful disorders such as ketosis. Similar studies can help to identify novel economically relevant traits for which selection in future is likely to be a priority (Miglior et al., 2017).

CONCLUSIONS

We estimated genetic parameters of a selection of blood traits (i.e., BHB_b, NEFA, and $urea_b$), indicators of cows' NEB, predicted from milk spectra in Italian Simmental cows. Moreover, we compared the performance of the offspring of top and bottom sires a posteriori. Although the prediction accuracy of the MIR models cannot be considered as optimal and does not allow for punctual determination of these hematic components, our findings suggest that predicted biomarkers have an exploitable genetic variability in the Italian Simmental population and enable the discrimination of animals with extreme genetic merit, e.g., top and bottom sires. However, results of the present study are based on a small sample of animals, and further studies are needed to confirm these findings. Although Simmental cattle tend to have a lower incidence of metabolic disorders compared with specialized dairy breeds, having access to easy-to-measure phenotypes routinely available on a large scale is crucial for monitoring NEB and ketosis across years and defining ad hoc breeding strategies to reduce or at least keep stable the resistance to metabolic diseases in future.

NOTES

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Nonstandard abbreviations used: BHB_b = blood BHB; BHB_m = milk BHB; MIR = mid-infrared; MY = milk yield; NEB = negative energy balance; NEFA = nonesterified fatty acid; *NS* = not significant; R^2_{CV} = coefficient of determination in cross-validation; R^2_V = coefficient of determination in external validation; r_a = pairwise genetic correlation; r_p = phenotypic correlation; $SET_{5–305}$ = entire dataset; $SET_{5–35}$ = subset containing only data from early lactation; $urea_b$ = blood urea; $urea_m$ = milk urea.

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